

HALSUA KANALA PUROLA, Finland

WMO: 027250

Lat: 63.446N

Lon: 24.444E

Elev: 153

StdP: 99.50

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.4	(c) -23.8	(d) -30.0	(e) 0.2	(f) -27.4	(g) -26.2	(h) 0.4	(i) -23.8	(j) 6.0	(k) -1.6	(l) 5.4	(m) -1.7	(n) 0.3	(o) 120	(p) 0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.2	(c) 26.9	(d) 18.2	(e) 24.6	(f) 16.9	(g) 22.7	(h) 16.1	(i) 19.5	(j) 24.4	(k) 18.2	(l) 22.9	(m) 17.1	(n) 21.2	(o) 2.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.0	21.3	16.6	12.0	20.1	15.3	11.0	18.8	56.4	24.2	52.1	23.0	48.7	21.1	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.3	(b) 4.6	(c) 4.2	(e) -28.6	(f) 28.5	(g) 4.8	(h) 2.7	(i) -32.0	(j) 30.4	(k) -34.8	(l) 32.0	(m) -37.5	(n) 33.5	(o) -41.0	(p) 35.5		
(a) 5.3	(b) 4.6	(c) 4.2	(e) -28.7	(f) 20.7	(g) 4.7	(h) 2.0	(i) -32.1	(j) 22.1	(k) -34.9	(l) 23.2	(m) -37.6	(n) 24.4	(o) -41.0	(p) 25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-9.5	-7.5	-3.7	2.0	8.4	12.2	15.4	13.5	9.1	2.8	-0.9	-4.9
	DBStd	9.54	7.19	7.18	5.32	3.83	4.51	3.81	3.52	3.31	3.47	3.84	4.55	6.17
	HDD10.0	2942	603	489	425	241	85	19	2	7	57	224	327	462
	HDD18.3	5567	862	723	683	489	309	188	104	152	277	481	577	720
	CDD10.0	437	0	0	0	1	34	84	170	117	29	1	0	0
	CDD18.3	21	0	0	0	0	1	3	13	4	0	0	0	0
	CDH23.3	324	0	0	0	0	32	60	188	44	0	0	0	0
	CDH26.7	57	0	0	0	0	2	9	41	6	0	0	0	0
Wind	WSAvg	1.9	1.7	2.0	2.1	1.9	2.1	1.9	1.6	1.7	1.8	2.0	2.0	2.0
Precipitation	PrecAvg	621	43	35	33	33	48	67	81	70	56	60	53	50
	PrecMax	817	89	69	55	74	94	115	154	160	134	111	99	105
	PrecMin	469	12	4	4	7	8	20	22	27	19	15	6	17
	PrecStd	93	19	16	13	17	24	29	28	33	34	27	23	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	5.1	8.3	18.3	26.5	28.0	29.7	27.5	20.2	13.8	9.5	5.2
		MCWB	2.4	3.2	3.8	9.8	15.1	17.5	19.5	19.9	15.2	11.8	8.6	4.5
	2%	DB	1.5	3.4	6.1	14.3	23.8	25.0	27.9	24.1	18.2	11.1	7.2	3.2
		MCWB	1.0	1.9	3.0	8.2	14.2	17.0	19.4	17.2	14.1	9.9	6.4	2.6
	5%	DB	0.5	1.8	4.2	11.0	21.2	22.5	25.3	22.0	16.5	9.8	5.9	2.1
		MCWB	0.1	1.0	1.8	6.1	13.1	15.3	18.1	16.1	12.9	8.8	5.6	1.6
	10%	DB	-0.3	0.7	2.7	8.9	18.0	20.3	23.3	20.1	14.9	8.4	4.6	1.2
		MCWB	-0.6	0.1	1.1	5.0	11.0	14.5	17.2	15.5	12.5	7.5	4.2	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	3.2	4.6	10.6	16.6	19.2	21.7	20.8	16.1	11.9	8.8	4.8
		MCDB	2.9	4.5	7.0	15.7	23.8	24.1	26.9	26.2	18.8	13.0	9.3	5.3
	2%	WB	1.0	2.1	3.2	8.7	15.1	17.8	20.1	18.5	14.8	10.4	6.6	2.7
		MCDB	1.2	3.0	5.2	13.7	21.4	23.0	25.0	22.4	17.1	11.0	7.1	3.0
	5%	WB	0.3	1.1	2.3	6.8	13.6	16.5	19.0	17.2	13.7	8.9	5.5	1.6
		MCDB	0.5	1.8	3.9	10.1	20.3	21.5	23.8	20.2	15.6	9.6	5.8	1.9
	10%	WB	-0.6	0.3	1.2	5.2	11.9	14.9	17.9	16.1	12.8	7.5	4.3	0.9
		MCDB	-0.3	0.8	2.4	8.3	17.8	19.1	22.5	19.1	14.4	8.3	4.6	1.2
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.1	8.8	9.8	12.3	11.3	11.2	10.4	8.7	6.0	3.8	4.6
		MCDBR	3.8	5.1	8.0	14.6	17.7	15.5	14.9	13.9	11.7	7.2	4.7	4.3
		MCWBR	3.7	4.3	5.7	9.0	9.5	8.3	7.5	7.8	7.8	5.9	4.5	4.1
	5% WB	MCDBR	3.8	5.0	7.3	13.3	16.2	13.2	13.3	11.2	9.6	6.5	4.5	4.2
		MCWBR	3.8	4.5	5.5	8.7	8.9	7.4	7.2	6.6	6.8	5.6	4.3	4.1
Clear-Sky Solar Irradiance	taub	0.221	0.282	0.309	0.350	0.345	0.346	0.367	0.352	0.328	0.308	0.374	0.356	
	taud	2.006	2.144	2.209	2.332	2.449	2.466	2.423	2.460	2.516	2.389	2.701	2.638	
	Ebn at Noon	384	637	779	829	870	875	841	823	775	621	357	210	
	Edh at Noon	30	63	89	99	96	97	99	87	68	51	27	17	
All-Sky Solar Radiation	RadAvg	0.17	0.73	2.22	3.73	4.85	5.28	4.98	3.79	2.17	0.84	0.21	0.07	
	RadStd	0.02	0.13	0.21	0.40	0.55	0.47	0.51	0.40	0.24	0.16	0.04	0.01	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (44 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page